Practical handoff and recovery

Saved simulation evidence | 26 October 2026

100 completed replicated runs: two plants, five families, SG/plain, five seeds.

3,840 paired E1 decisions: 3,200 polymer + all 640 column, combined 320/320.

24 figures; complete raw-data checks, provenance and an unchanged prior assessment.

Critic versus simple threshold, handoff episodes 14-20 (single-seed E2):

- Polymer: output RMSEs 37.0% / 46.3% lower; input activity 76.6% lower.
- Column: output RMSEs 6.3% / 41.1% lower; input activity 61.9% lower.

Reward recovery after release 14, critic versus threshold: polymer 1 vs 25; column 5 vs 37.

Both-output RMSE recovery: polymer 1 vs 25; column 37 vs 22 (threshold wins).

Recovery requires three consecutive qualifying episodes relative to matched MPC.

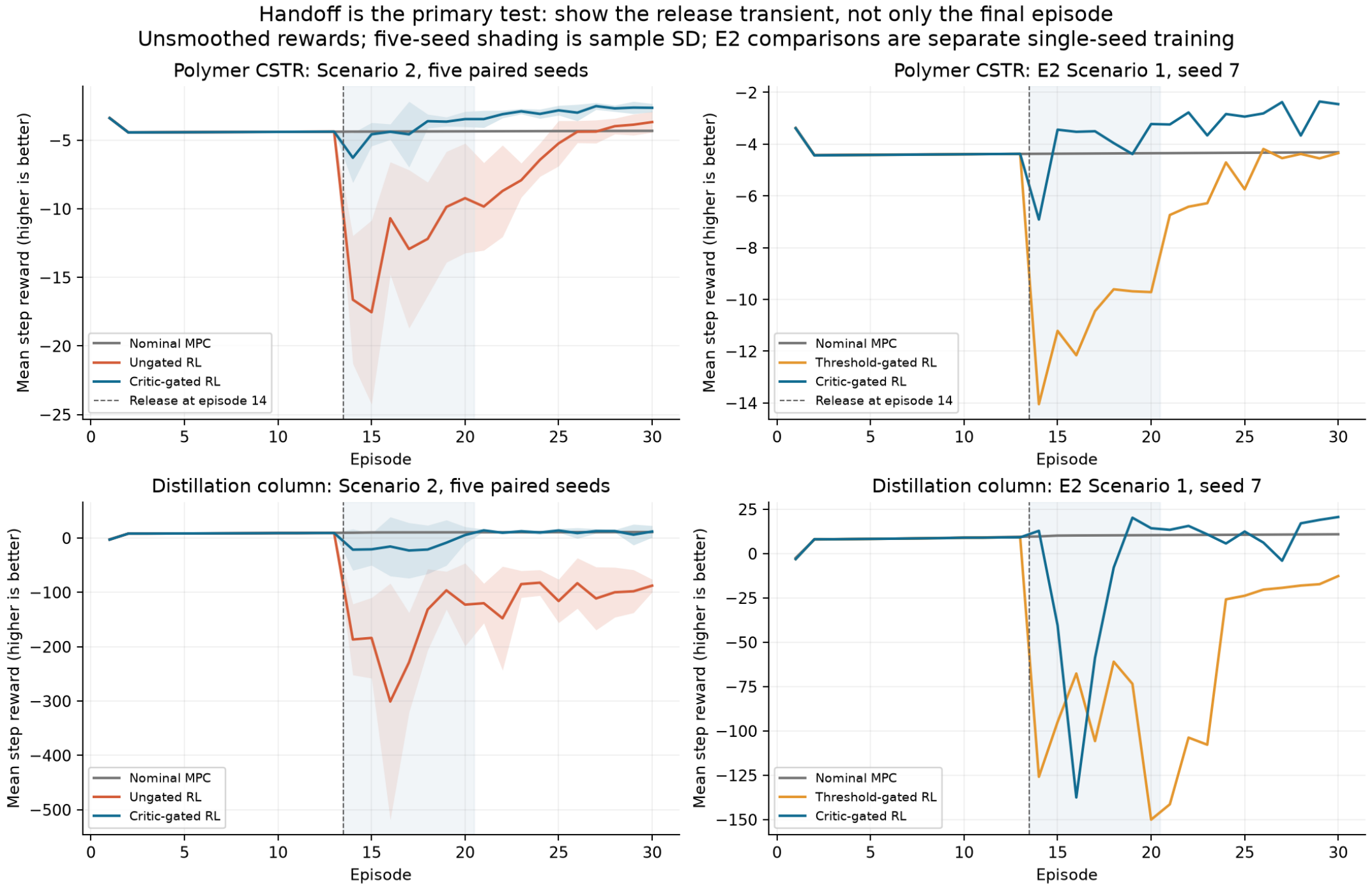
Full combined E1: 100/200 accepted policy actions have lower paired 200-step return.

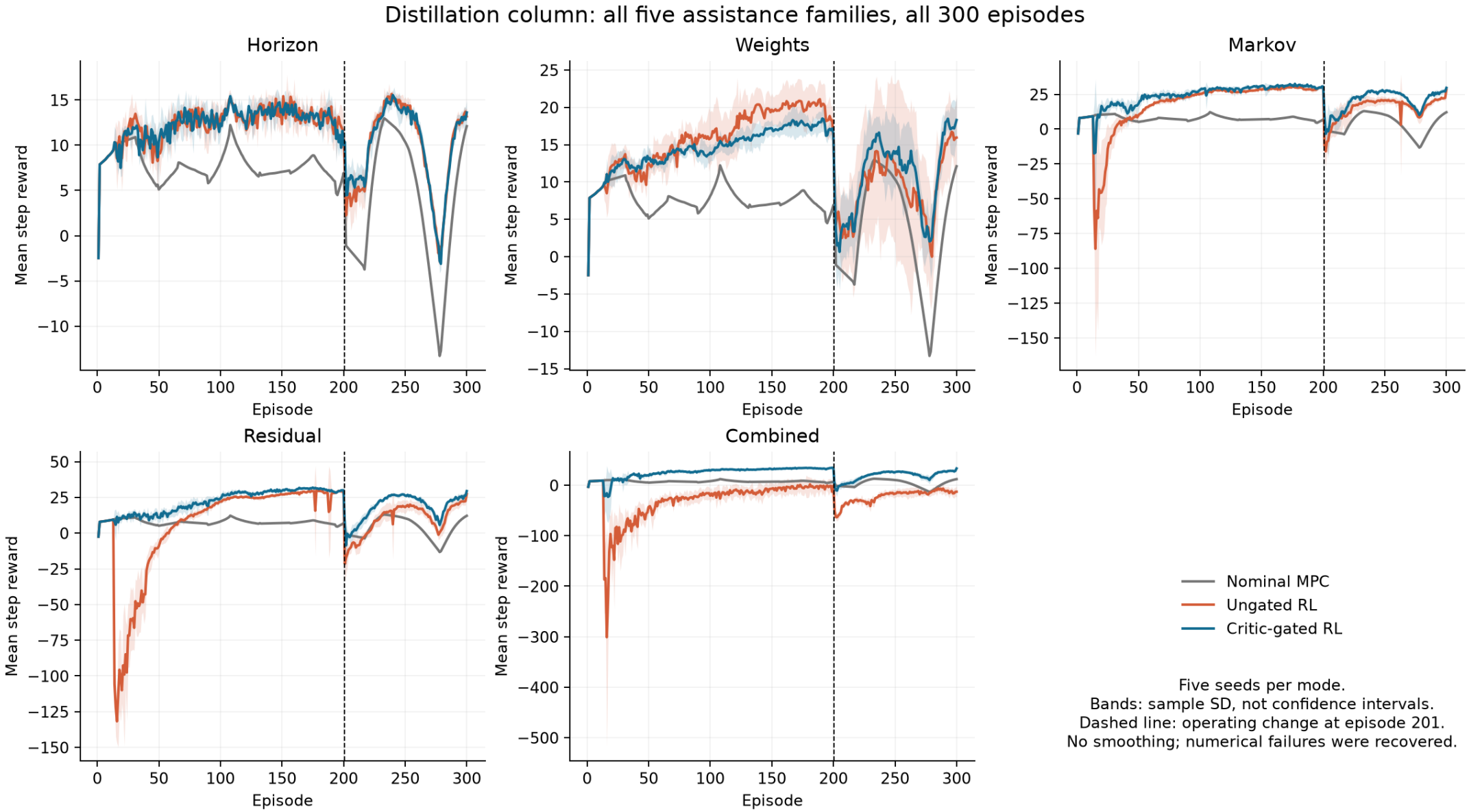
This supports practical handoff mitigation, not reliable critic ordering or formal safety.

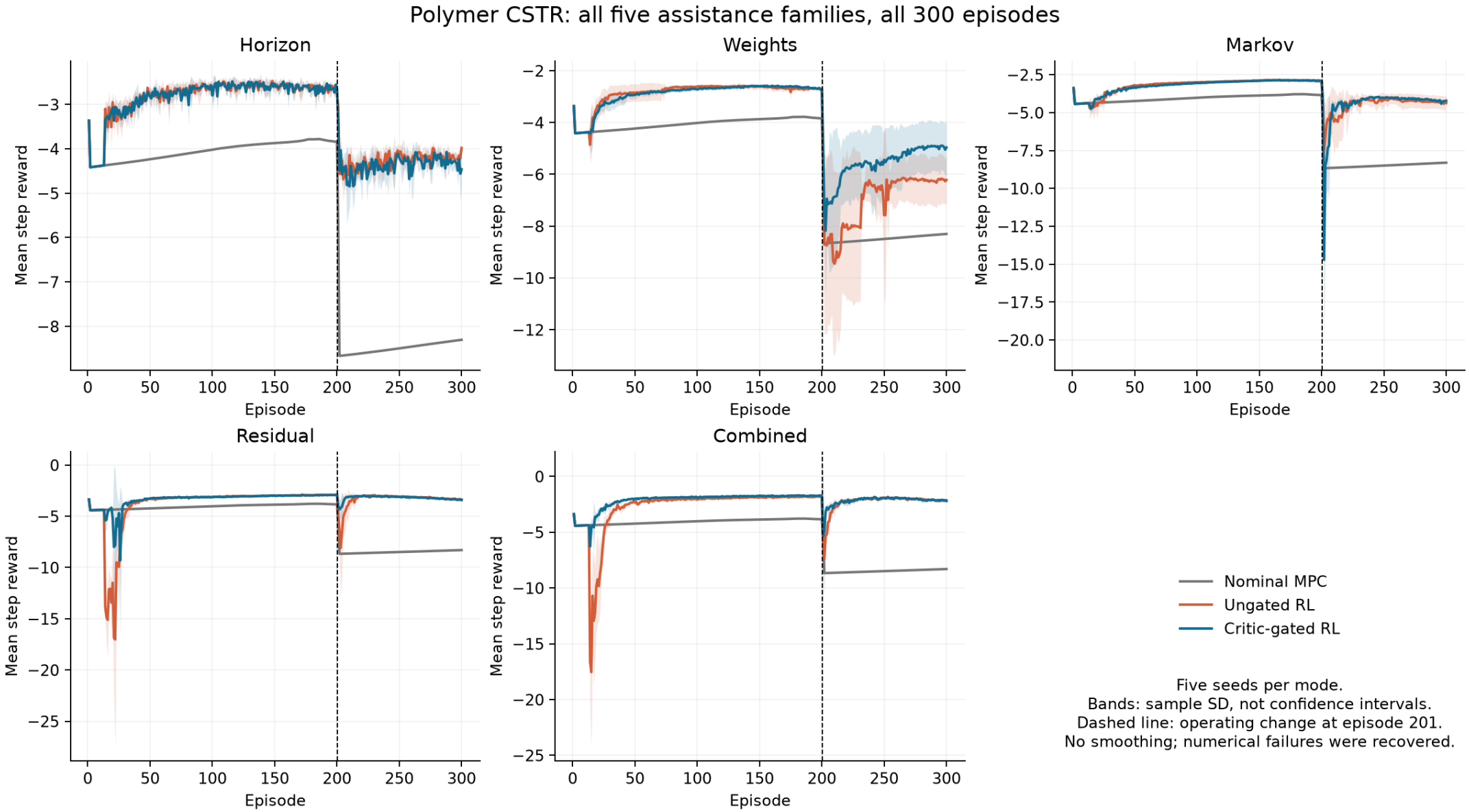
Steady-state equivalence is partly supported; adverse decisions remain.

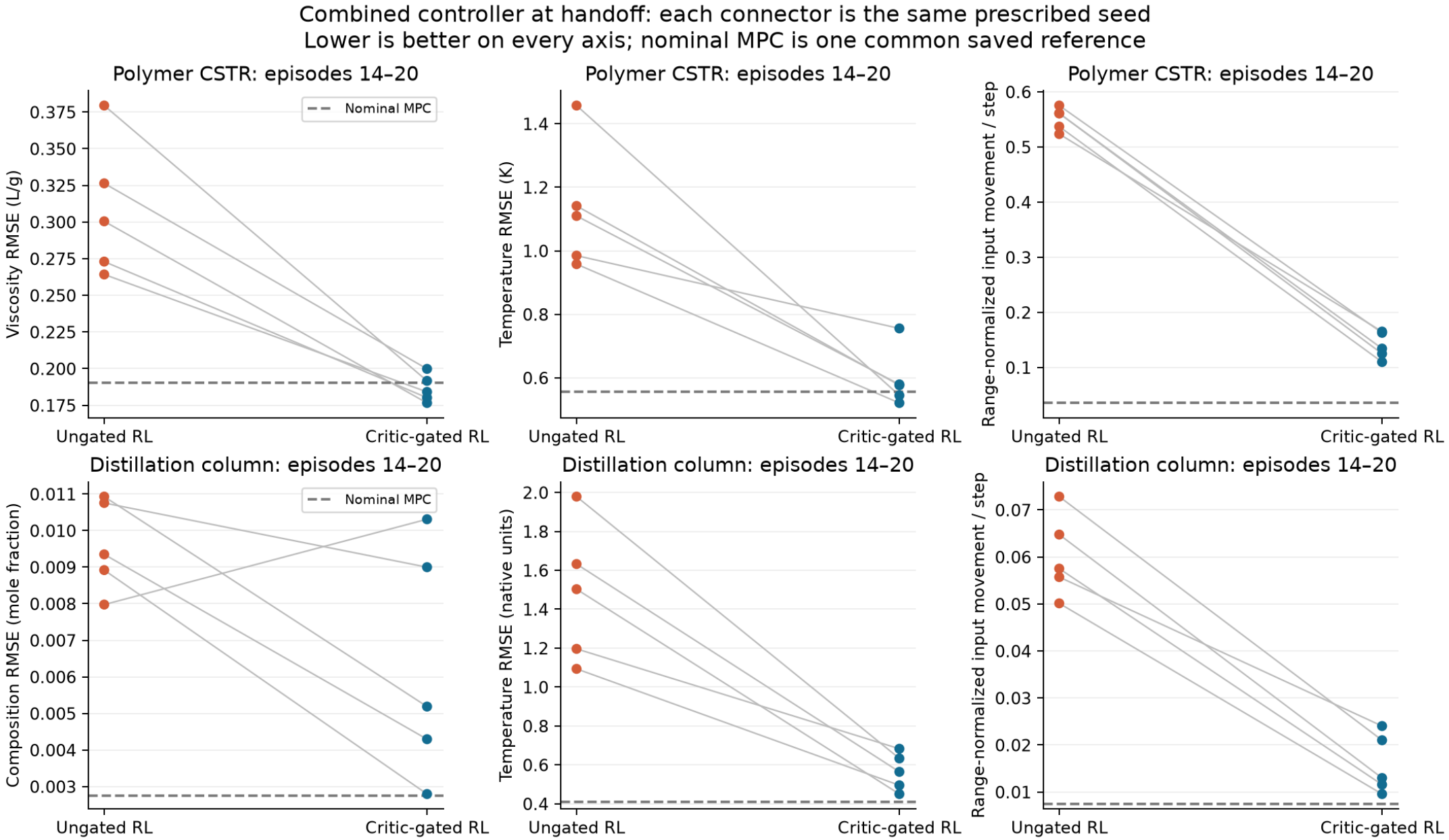
Reviewer preparation can begin. Unseen references, E4/E5 and analytical/literature dependencies remain distinct. No comments answered or manuscript changes.

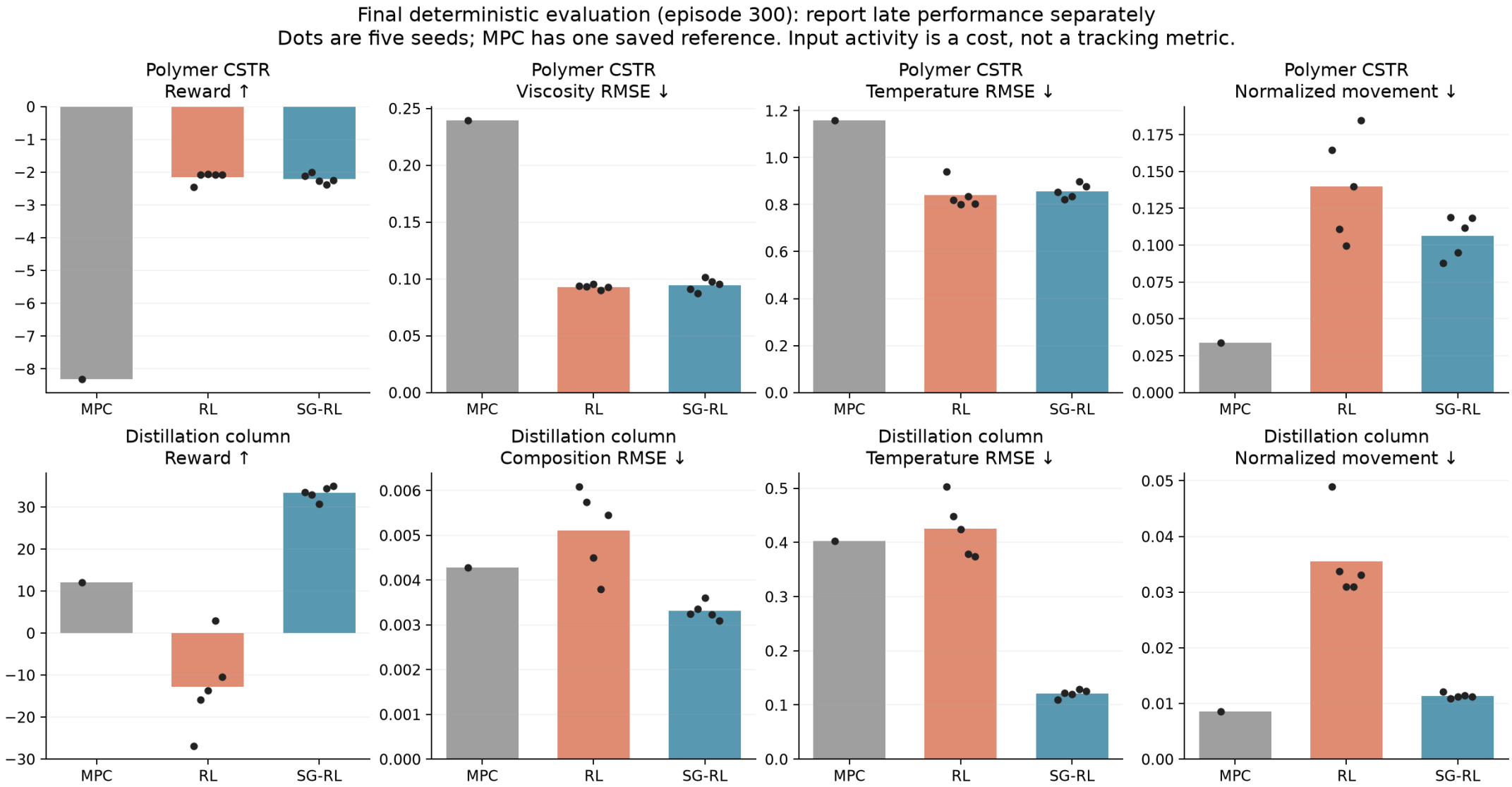
Read ASSESSMENT.md and REVIEWER_READINESS.md for definitions and limitations.

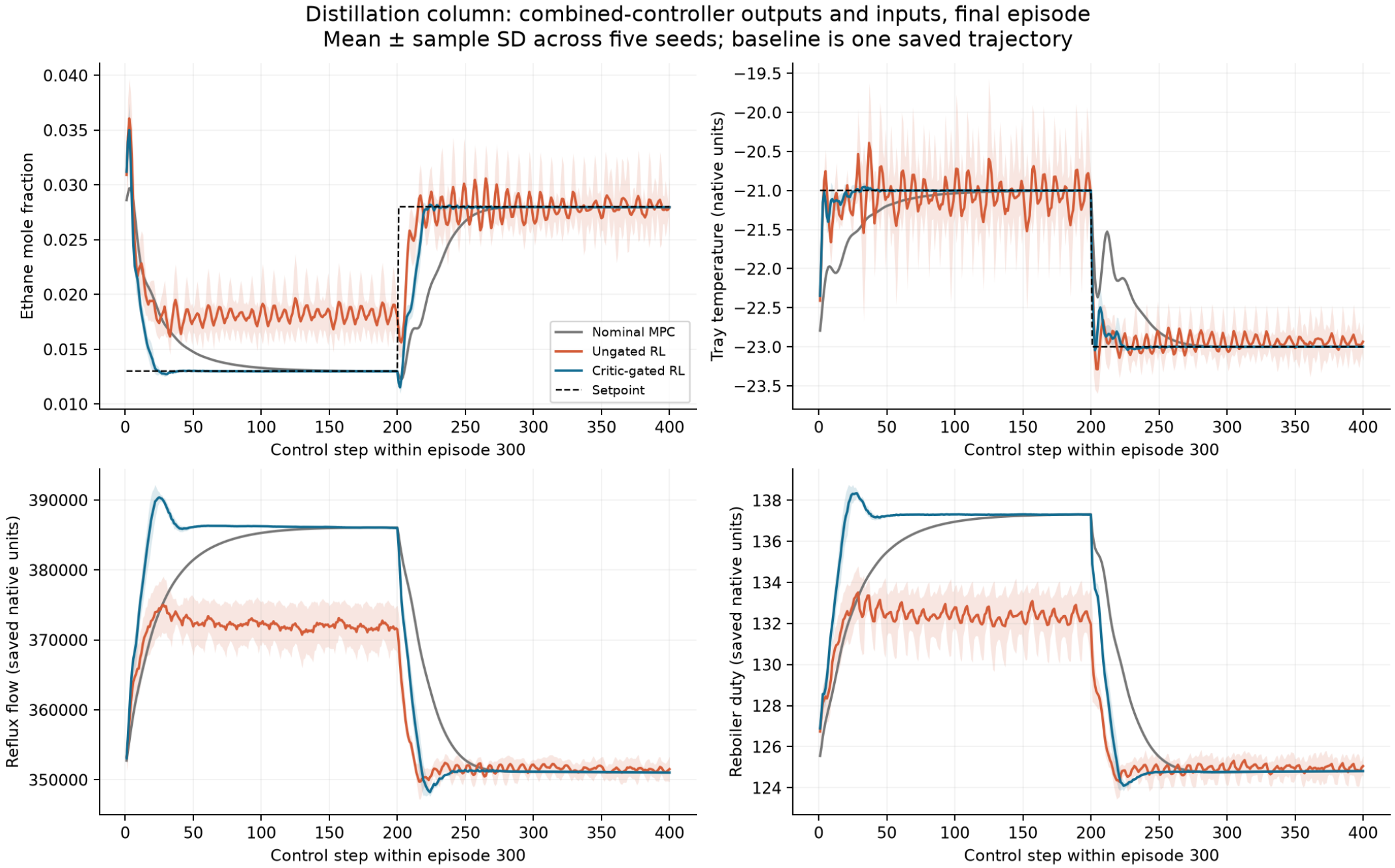

Handoff is the primary test: show the release transient, not only the final episode
Unsmoothed rewards; five-seed shading is sample SD; E2 comparisons are separate single-seed training
Polymer CSTR: Scenario 2, five paired seeds
Polymer CSTR: E2 Scenario 1, seed 7
Distillation column: Scenario 2, five paired seeds
Distillation column: E2 Scenario 1, seed 7
Mean step reward (higher is better)
Episode
Nominal MPC
Ungated RL
Threshold-gated RL
Critic-gated RL
Release at episode 14

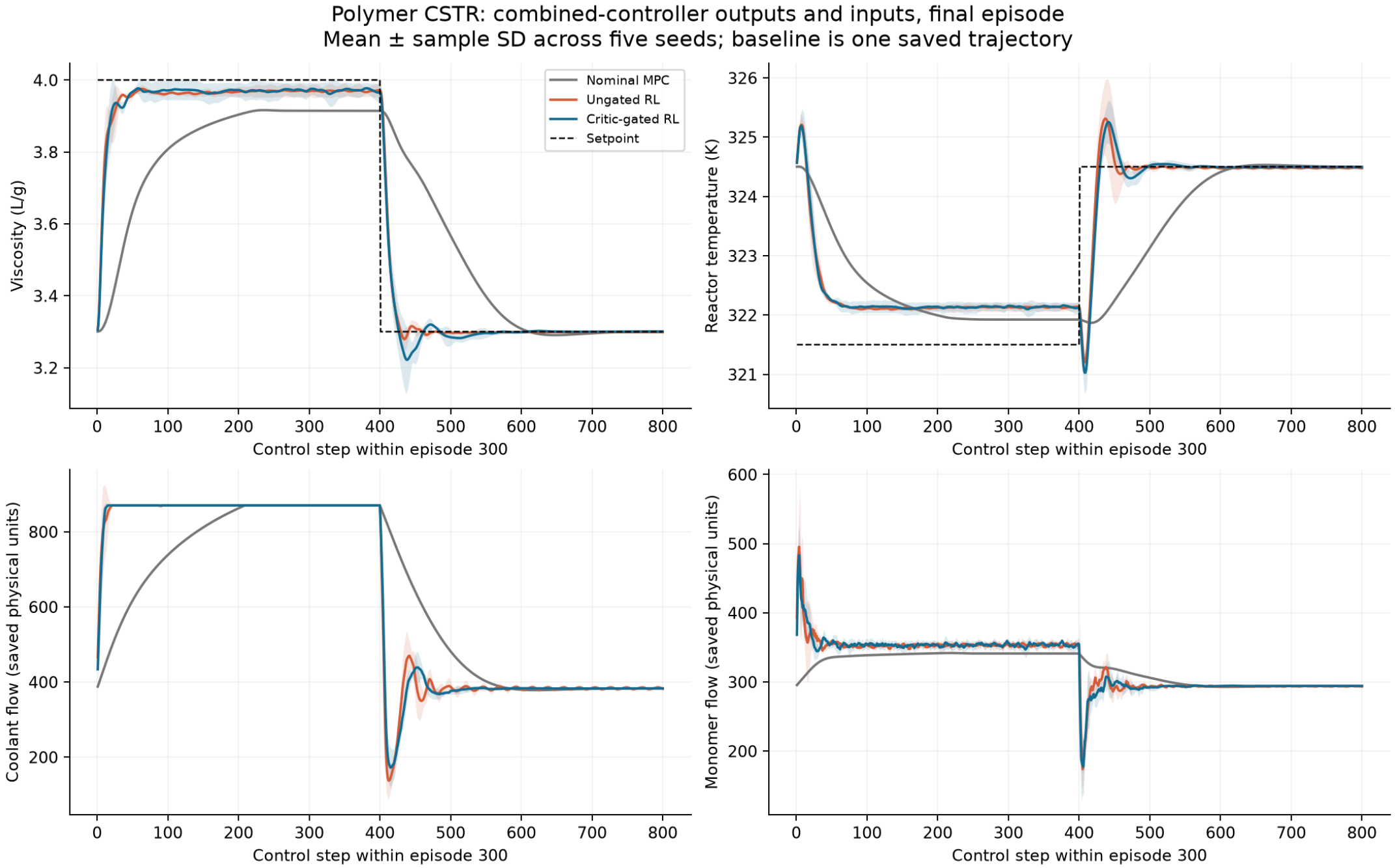

Distillation column: all five assistance families, all 300 episodes
Horizon
Weights
Markov
Residual
Combined
Mean step reward
Episode
Nominal MPC
Ungated RL
Critic-gated RL
Five seeds per mode.
Bands: sample SD, not confidence intervals.
Dashed line: operating change at episode 201.
No smoothing; numerical failures were recovered.

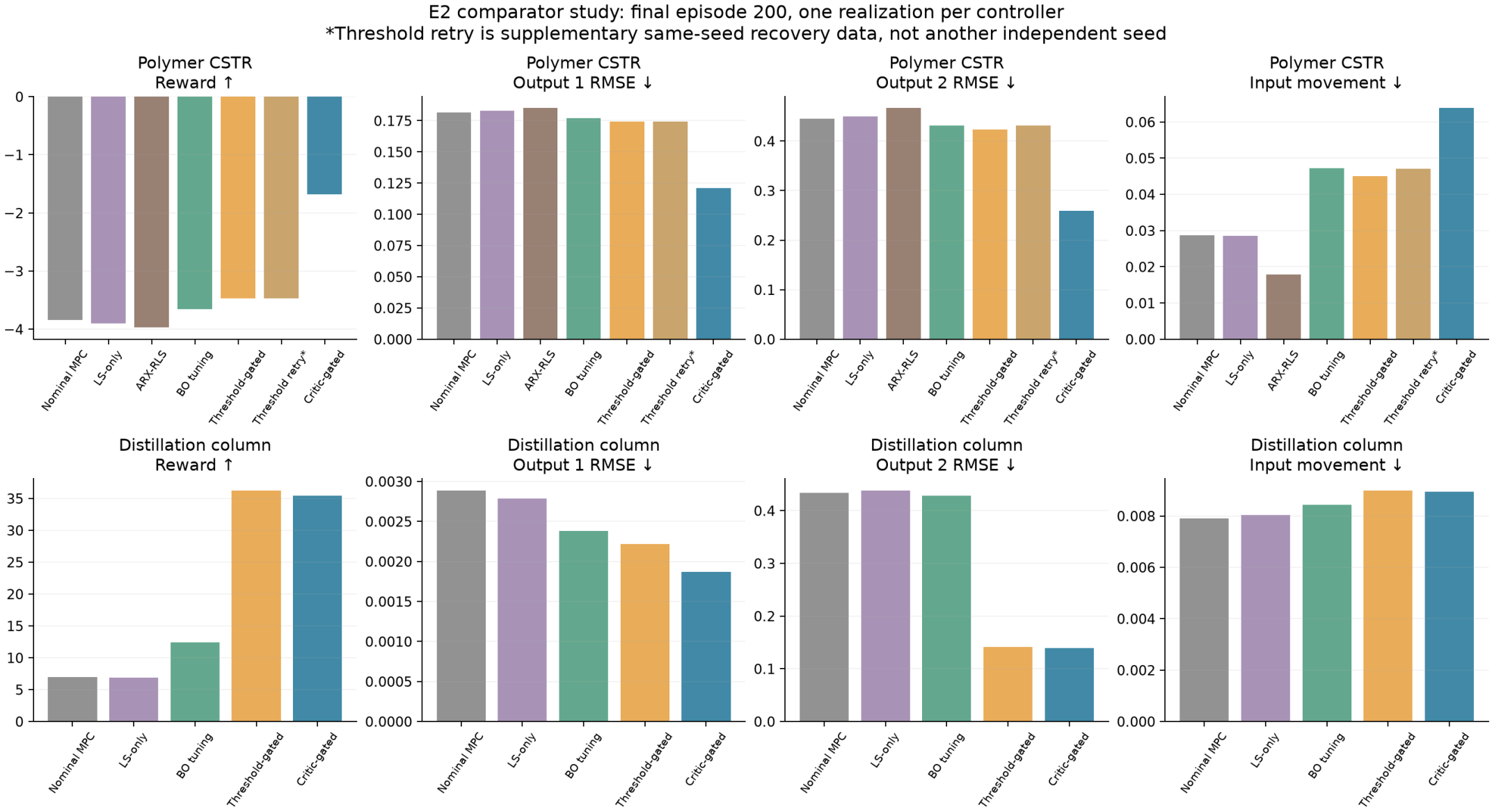

Polymer CSTR: all five assistance families, all 300 episodes
Horizon
Weights
Markov
Residual
Combined
Mean step reward
Episode
Nominal MPC
Ungated RL
Critic-gated RL
Five seeds per mode.
Bands: sample SD, not confidence intervals.
Dashed line: operating change at episode 201.
No smoothing; numerical failures were recovered.

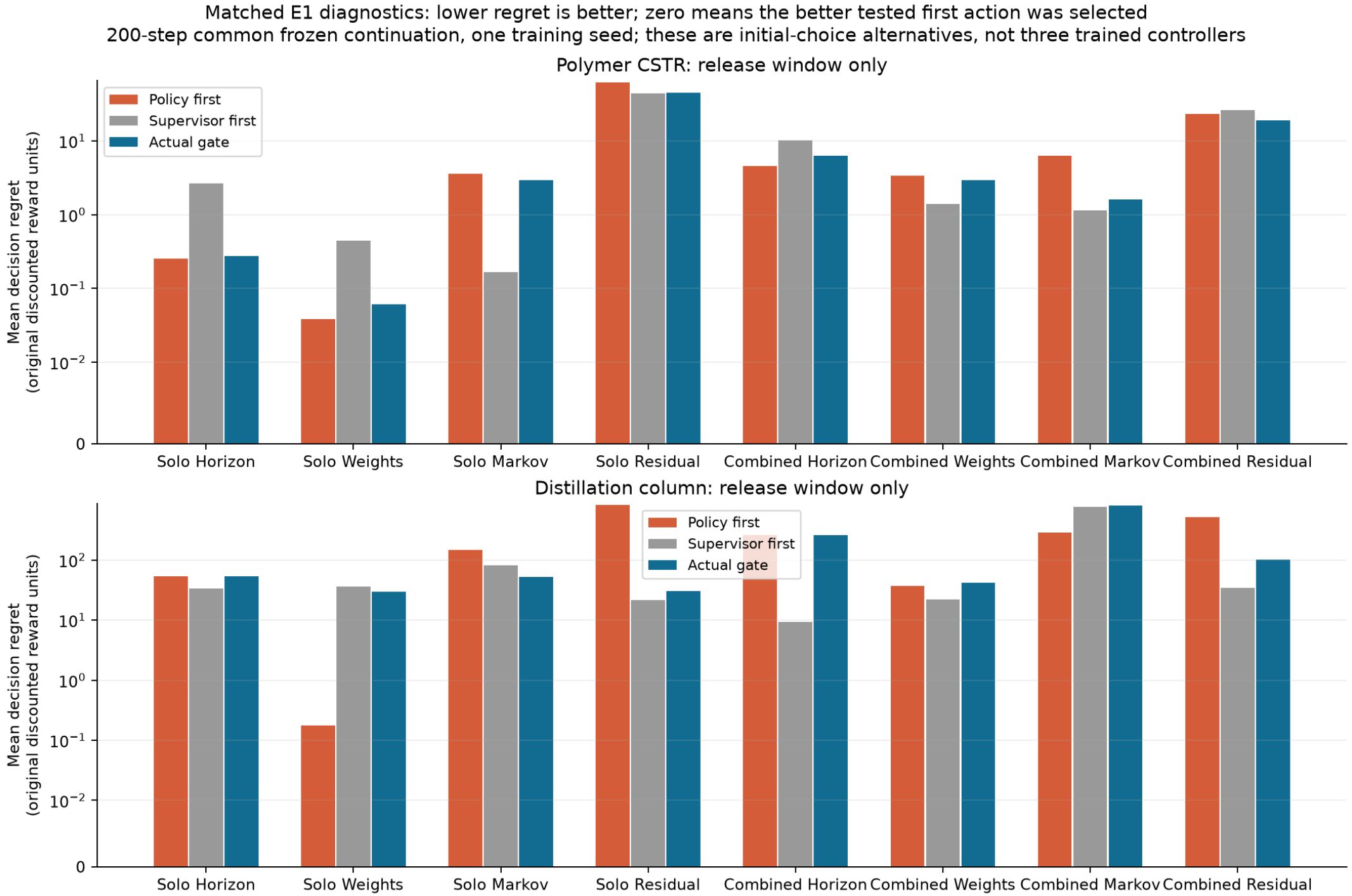

Combined controller at handoff: each connector is the same prescribed seed
Lower is better on every axis; nominal MPC is one common saved reference
Polymer CSTR: episodes 14–20
Viscosity RMSE (L/g)
Nominal MPC
Ungated RL
Critic-gated RL
Polymer CSTR: episodes 14–20
Temperature RMSE (K)
Ungated RL
Critic-gated RL
Polymer CSTR: episodes 14–20
Range-normalized input movement / step
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Composition RMSE (mole fraction)
Nominal MPC
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Temperature RMSE (native units)
Ungated RL
Critic-gated RL
Distillation column: episodes 14–20
Range-normalized input movement / step
Ungated RL
Critic-gated RL

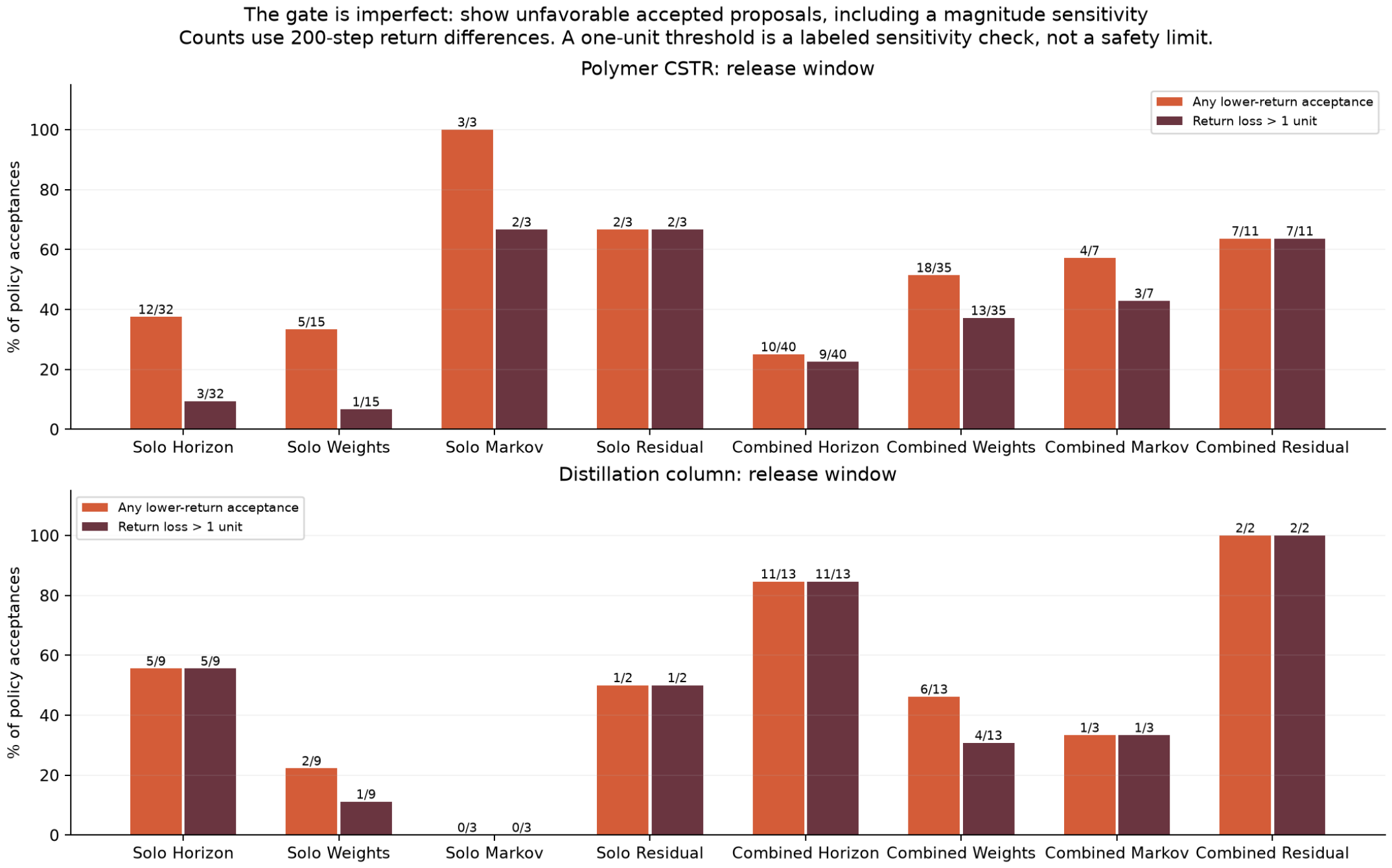

Final deterministic evaluation (episode 300): report late performance separately
Dots are five seeds; MPC has one saved reference. Input activity is a cost, not a tracking metric.
Polymer CSTR
Reward ↑
Polymer CSTR
Viscosity RMSE ↓
Polymer CSTR
Temperature RMSE ↓
Polymer CSTR
Normalized movement ↓
Distillation column
Reward ↑
Distillation column
Composition RMSE ↓
Distillation column
Temperature RMSE ↓
Distillation column
Normalized movement ↓
MPC
RL
SG-RL


Distillation column: combined-controller outputs and inputs, final episode
Mean ± sample SD across five seeds; baseline is one saved trajectory
Ethane mole fraction
Tray temperature (native units)
Reflux flow (saved native units)
Reboiler duty (saved native units)
Control step within episode 300
Nominal MPC
Ungated RL
Critic-gated RL
Setpoint


Polymer CSTR: combined-controller outputs and inputs, final episode
Mean ± sample SD across five seeds; baseline is one saved trajectory
Nominal MPC
Ungated RL
Critic-gated RL
Setpoint
Viscosity (L/g)
Reactor temperature (K)
Coolant flow (saved physical units)
Monomer flow (saved physical units)
Control step within episode 300


E2 comparator study: final episode 200, one realization per controller
*Threshold retry is supplementary same-seed recovery data, not another independent seed
Polymer CSTR
Reward ↑
Polymer CSTR
Output 1 RMSE ↓
Polymer CSTR
Output 2 RMSE ↓
Polymer CSTR
Input movement ↓
Distillation column
Reward ↑
Distillation column
Output 1 RMSE ↓
Distillation column
Output 2 RMSE ↓
Distillation column
Input movement ↓
Nominal MPC
LS-only
ARX-RLS
BO tuning
Threshold-gated
Threshold retry*
Critic-gated

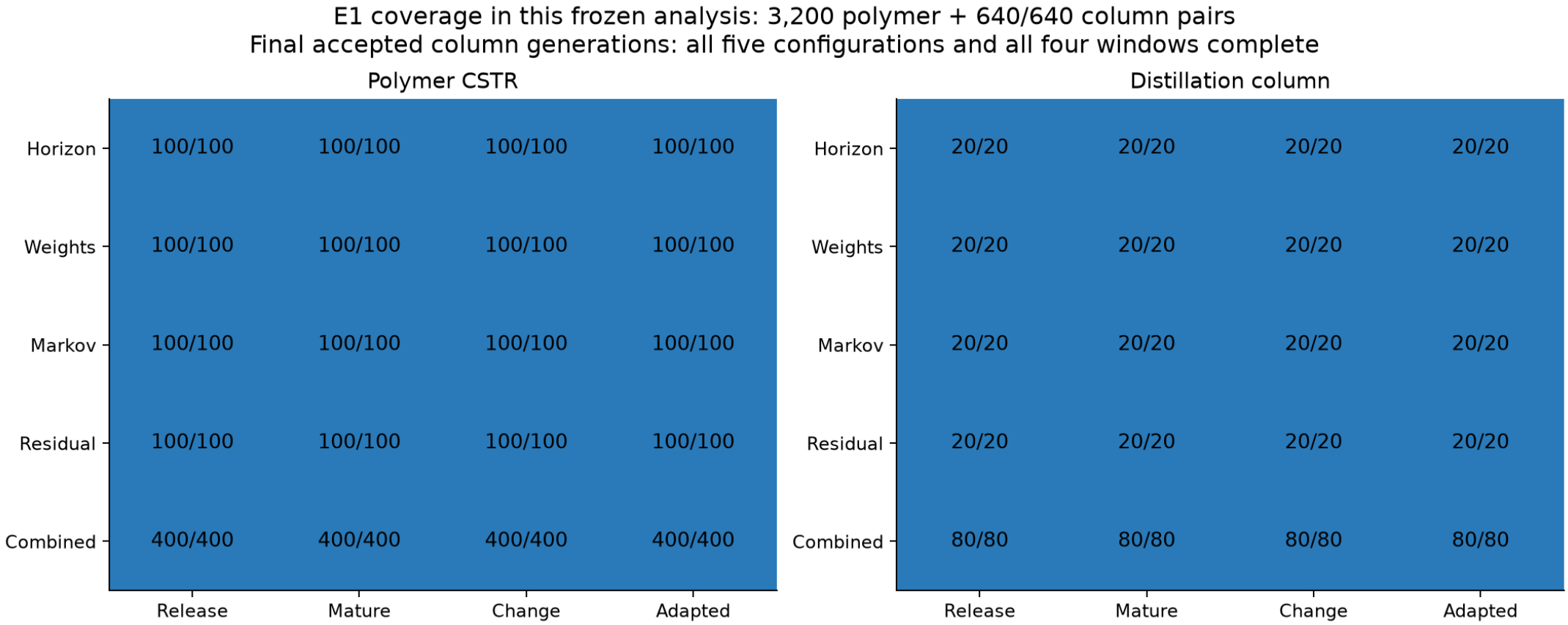

Matched E1 diagnostics: lower regret is better; zero means the better tested first action was selected
200-step common frozen continuation, one training seed; these are initial-choice alternatives, not three trained controllers
Polymer CSTR: release window only
Policy first
Supervisor first
Actual gate
Mean decision regret
(original discounted reward units)
10^{1}
10^{0}
10^{-1}
10^{-2}
0
Solo Horizon
Solo Weights
Solo Markov
Solo Residual
Combined Horizon
Combined Weights
Combined Markov
Combined Residual
Distillation column: release window only
Policy first
Supervisor first
Actual gate
Mean decision regret
(original discounted reward units)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0
Solo Horizon
Solo Weights
Solo Markov
Solo Residual
Combined Horizon
Combined Weights
Combined Markov
Combined Residual


The gate is imperfect: show unfavorable accepted proposals, including a magnitude sensitivity
Counts use 200-step return differences. A one-unit threshold is a labeled sensitivity check, not a safety limit.
Polymer CSTR: release window
Any lower-return acceptance
Return loss > 1 unit
% of policy acceptances
0
20
40
60
80
100
12/32
3/32
5/15
1/15
3/3
2/3
2/3
2/3
10/40
9/40
18/35
13/35
4/7
3/7
7/11
7/11
Solo Horizon
Solo Weights
Solo Markov
Solo Residual
Combined Horizon
Combined Weights
Combined Markov
Combined Residual
Distillation column: release window
Any lower-return acceptance
Return loss > 1 unit
% of policy acceptances
0
20
40
60
80
100
5/9
5/9
2/9
1/9
0/3
0/3
1/2
1/2
11/13
11/13
6/13
4/13
1/3
1/3
2/2
2/2
Solo Horizon
Solo Weights
Solo Markov
Solo Residual
Combined Horizon
Combined Weights
Combined Markov
Combined Residual

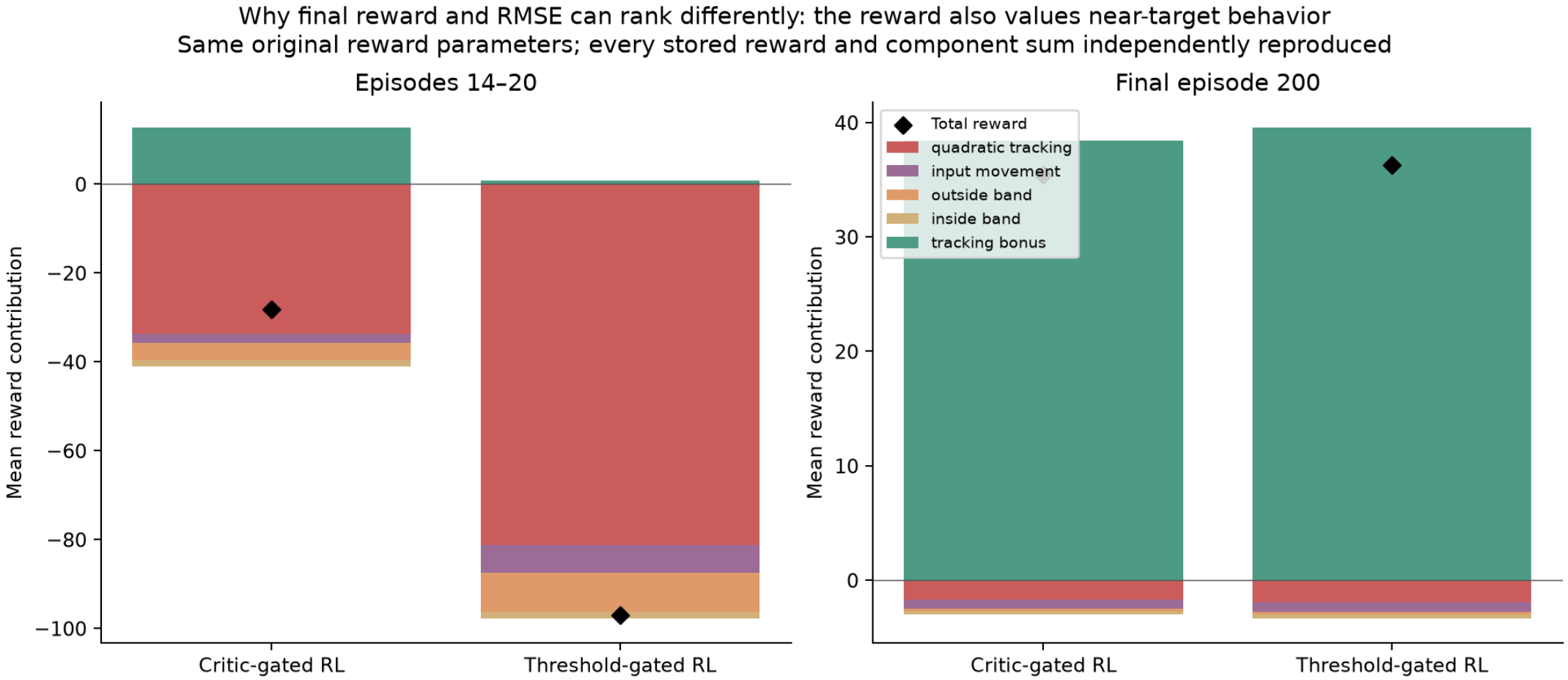
E3 held-out prediction ratios

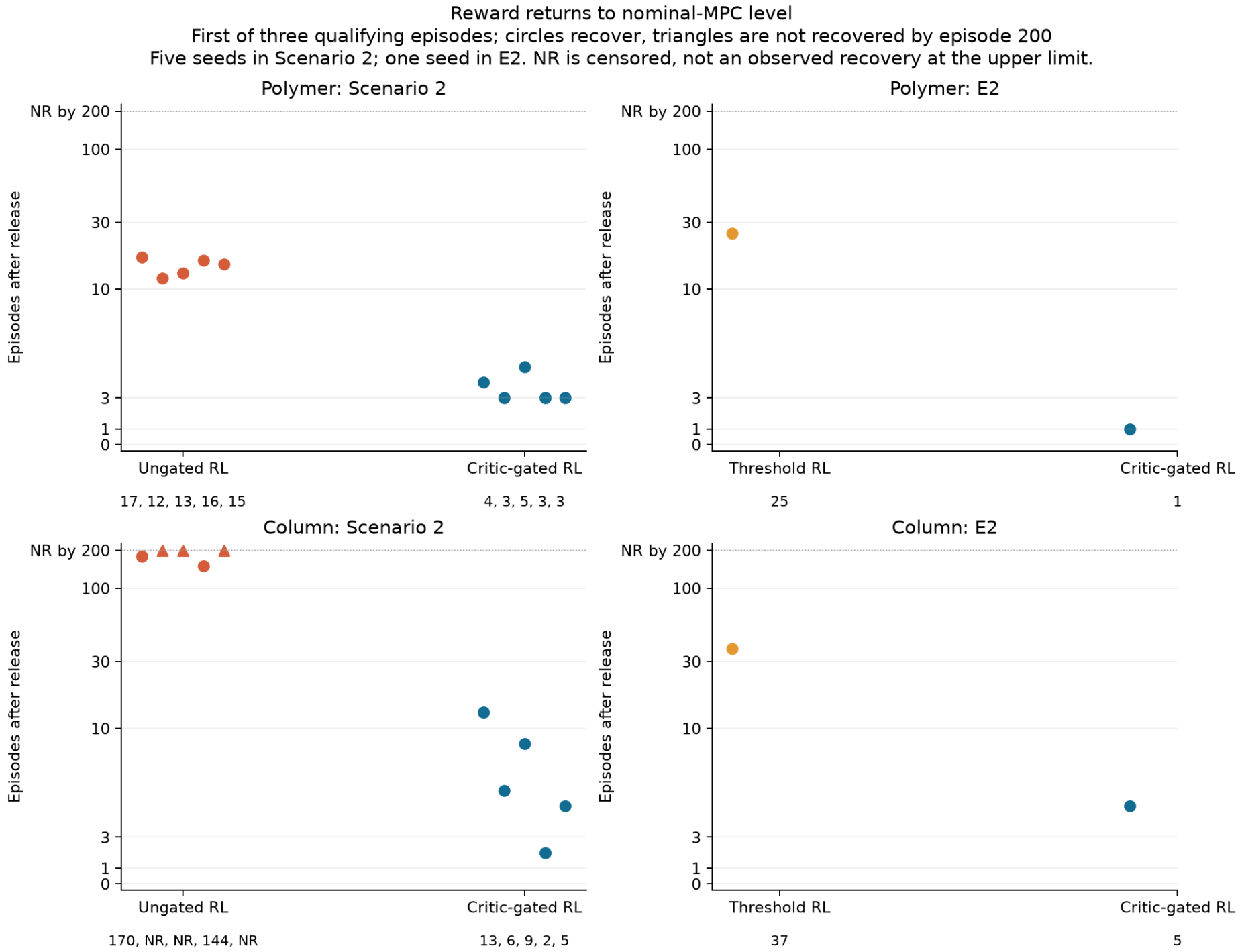


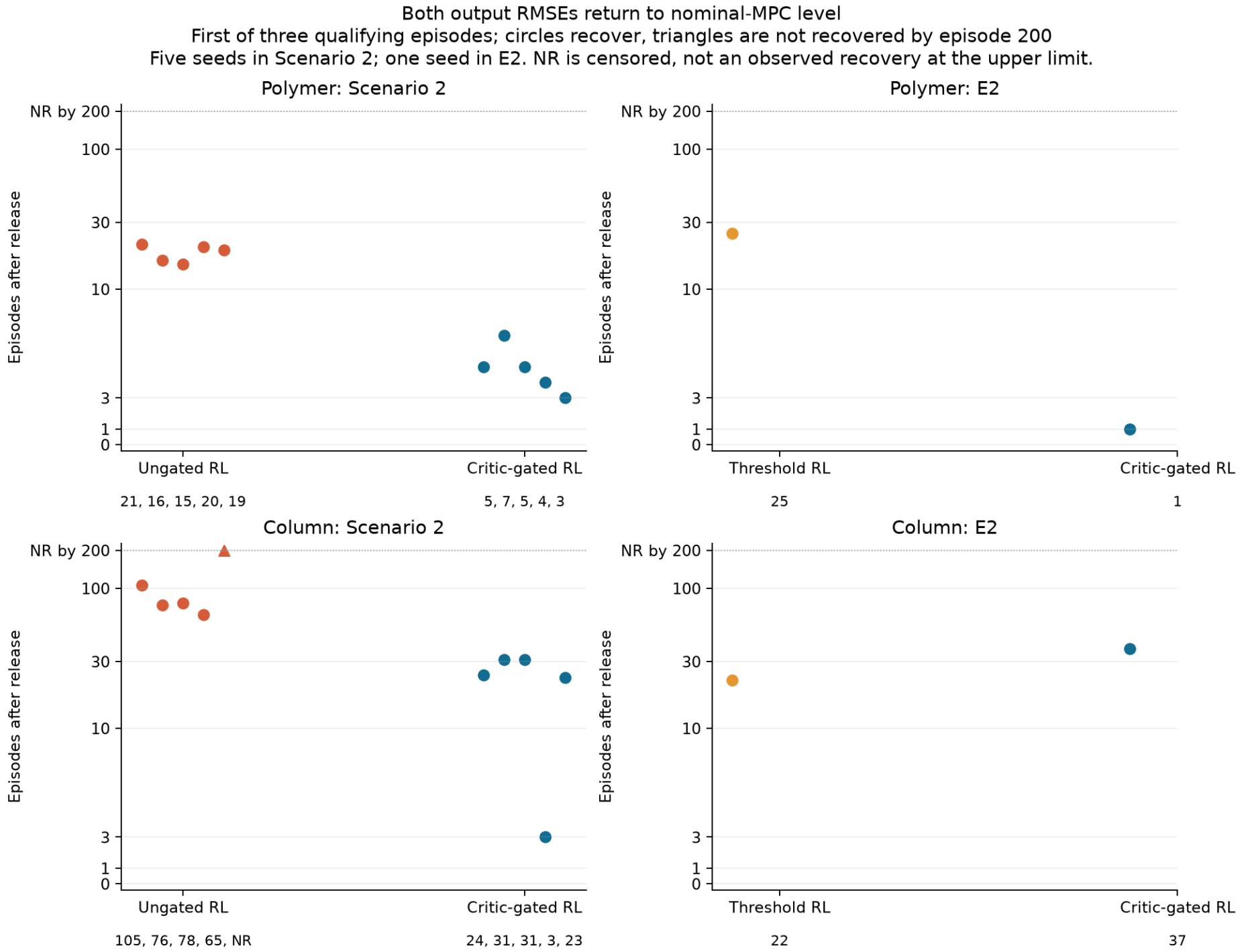
Complete polymer and column E1 coverage

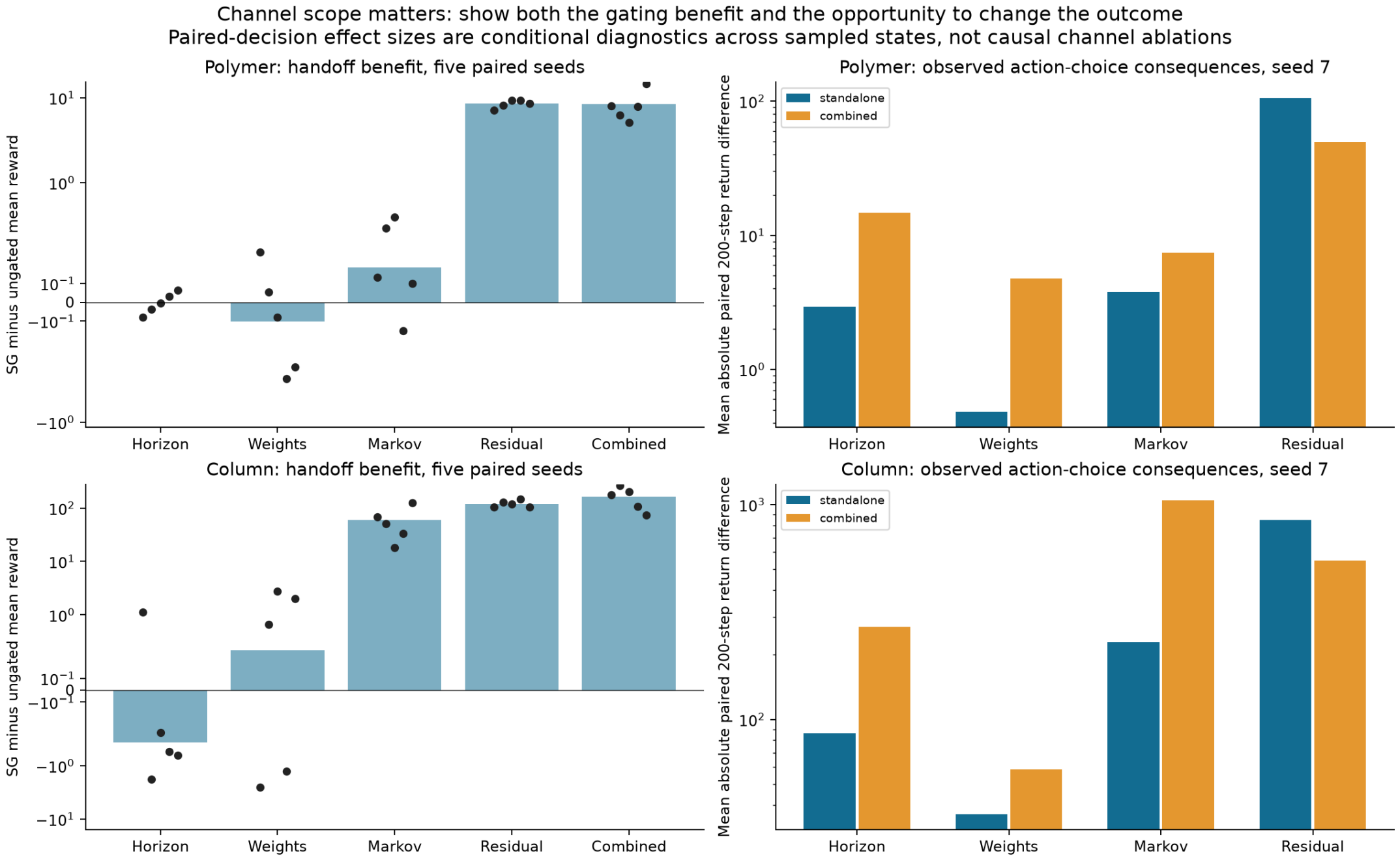


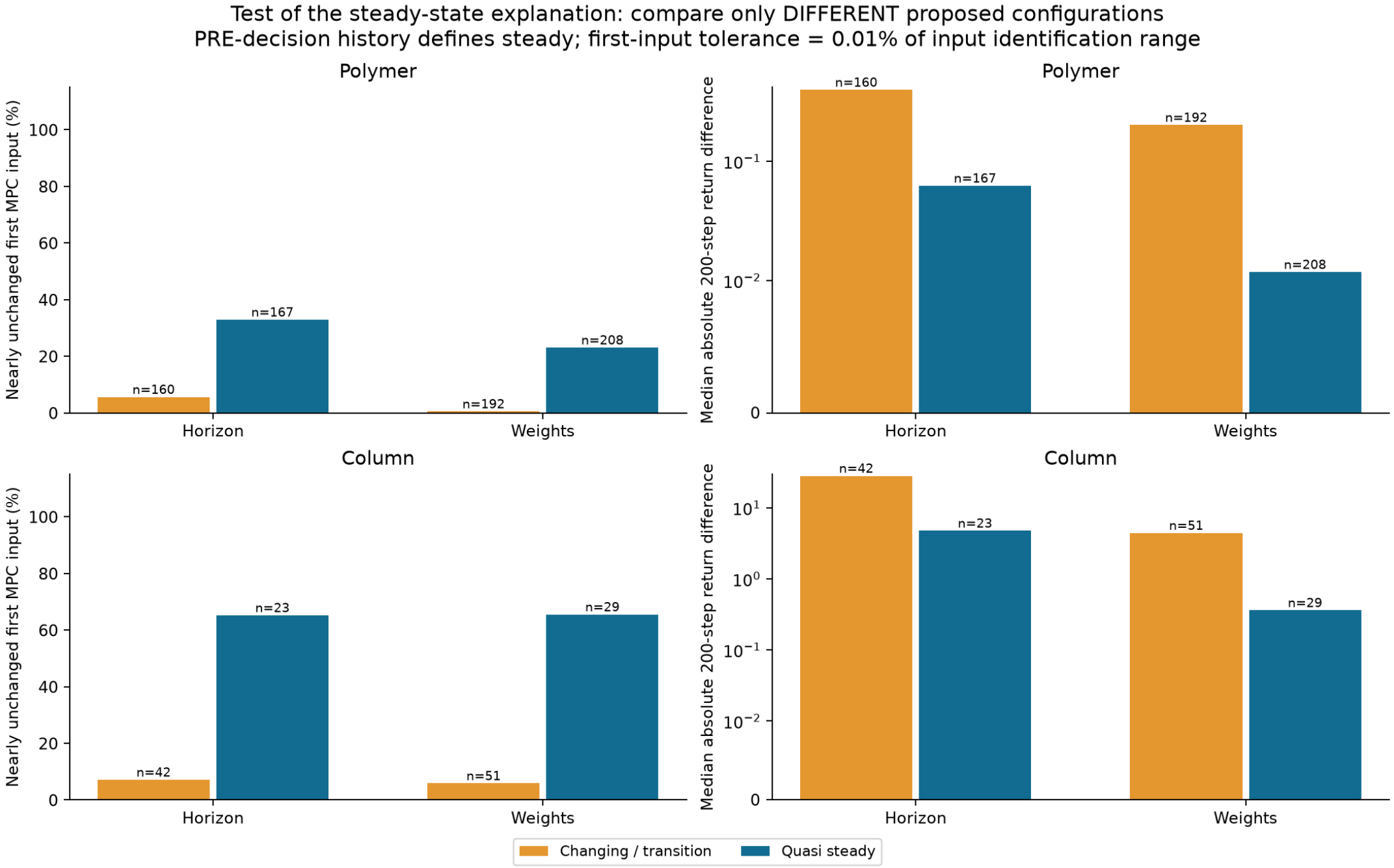
Why scalar reward and output RMSE can rank differently

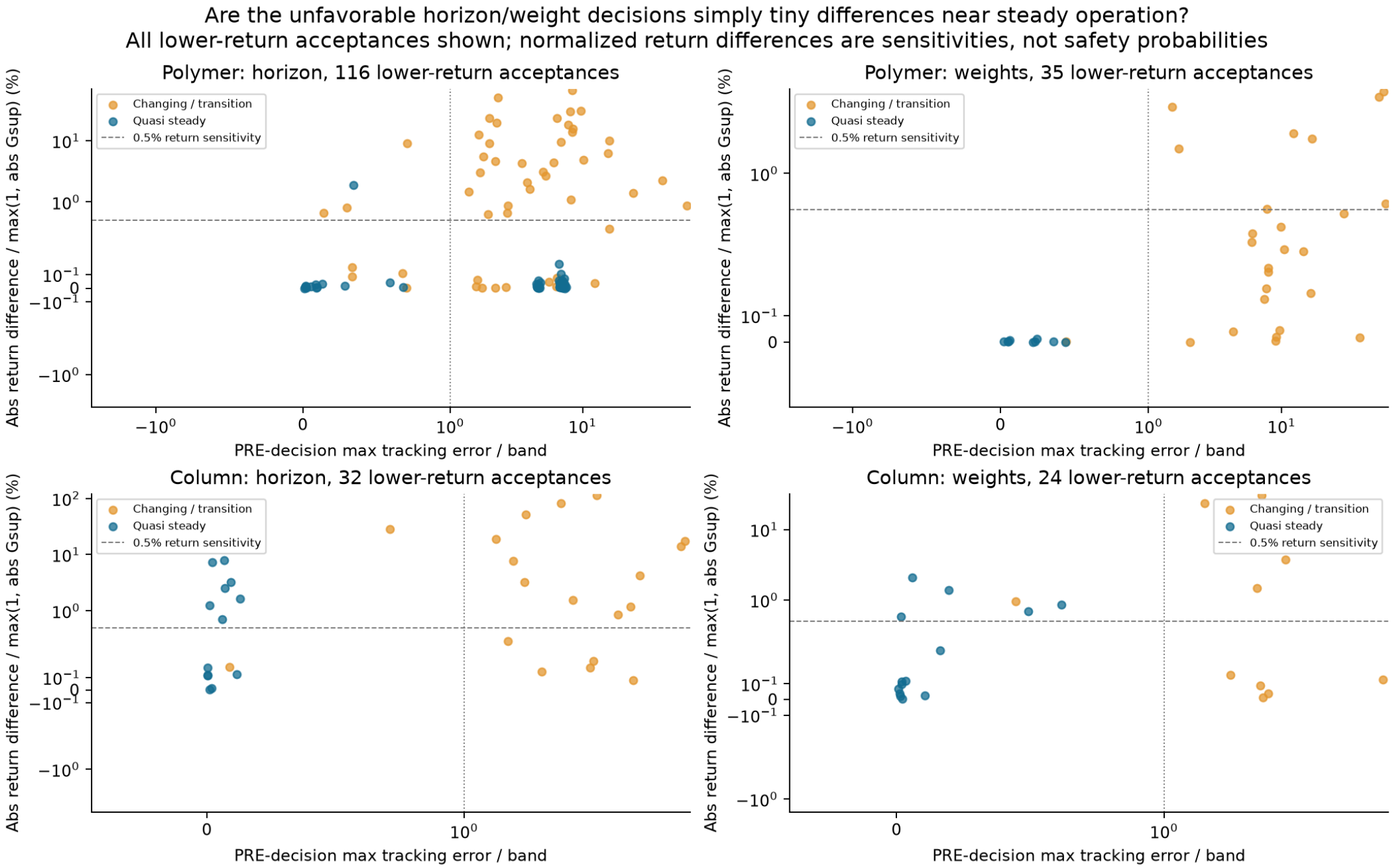


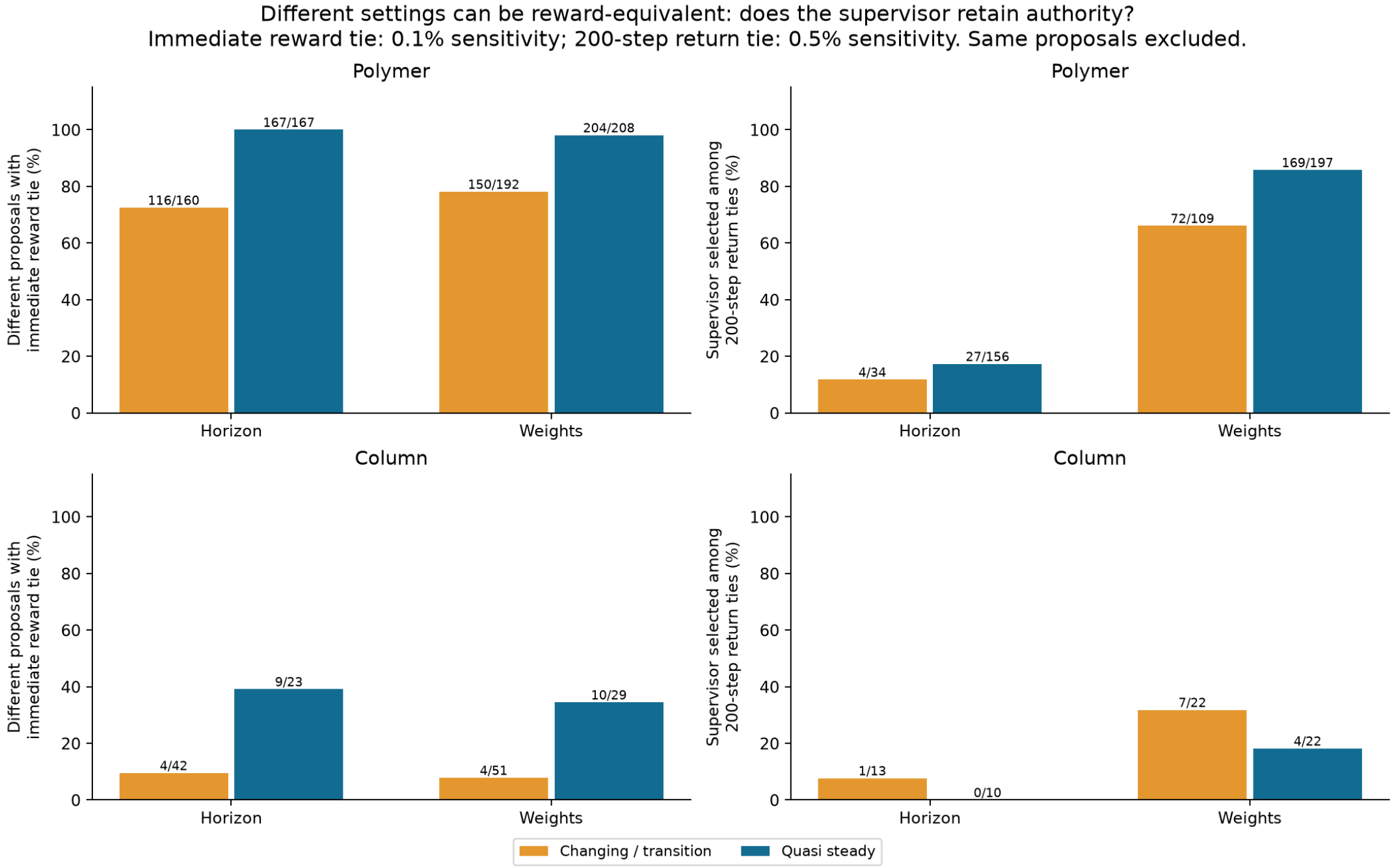

Reward returns to nominal-MPC level
First of three qualifying episodes; circles recover, triangles are not recovered by episode 200
Five seeds in Scenario 2; one seed in E2. NR is censored, not an observed recovery at the upper limit.
Polymer: Scenario 2
Polymer: E2
Column: Scenario 2
Column: E2
Episodes after release
NR by 200
100
30
10
3
1
0
Ungated RL
Critic-gated RL
Threshold RL
17, 12, 13, 16, 15
4, 3, 5, 3, 3
25
1
170, NR, NR, 144, NR
13, 6, 9, 2, 5
37
5

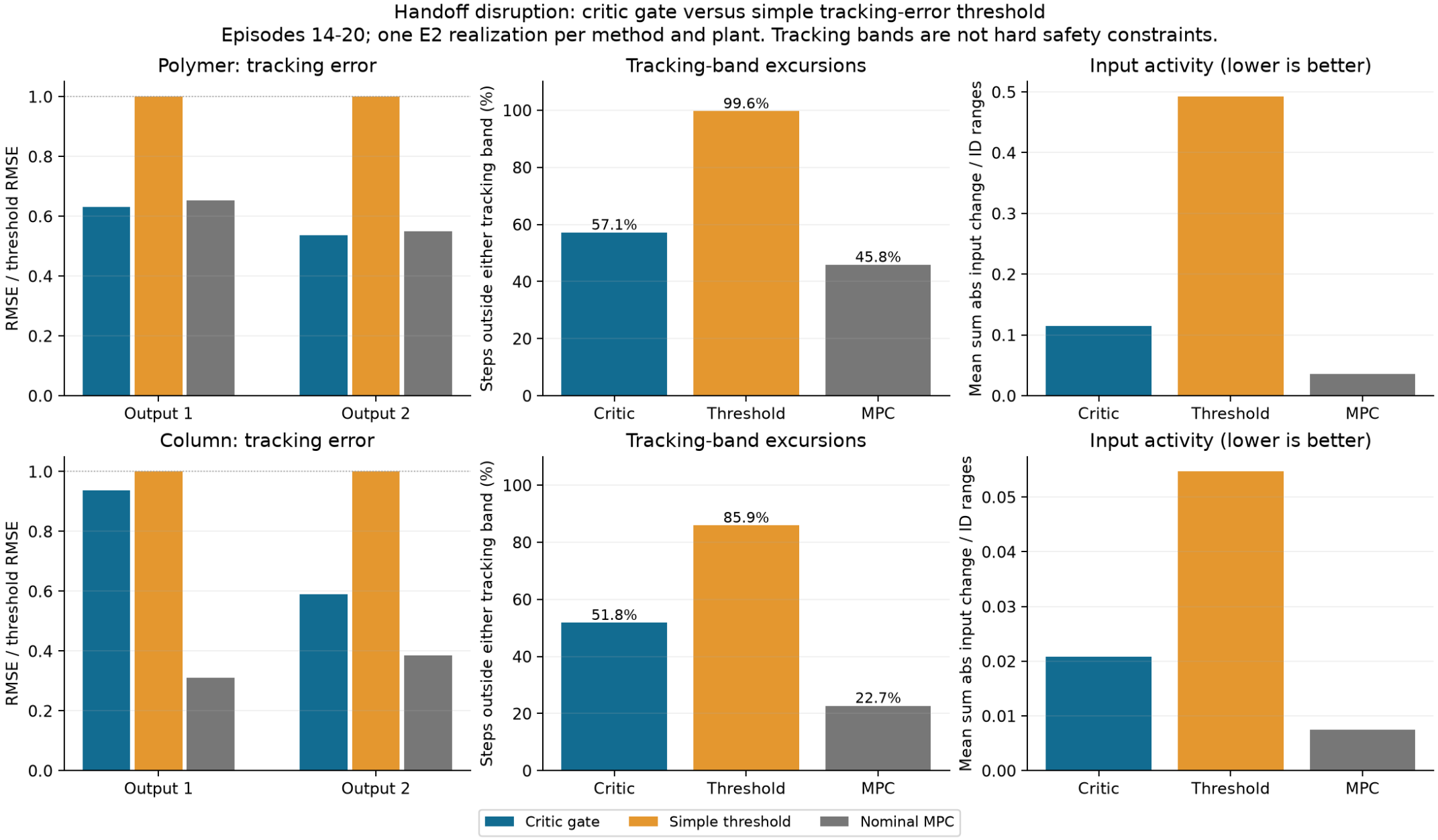

Both output RMSEs return to nominal-MPC level
First of three qualifying episodes; circles recover, triangles are not recovered by episode 200
Five seeds in Scenario 2; one seed in E2. NR is censored, not an observed recovery at the upper limit.
Polymer: Scenario 2
Polymer: E2
Column: Scenario 2
Column: E2
Episodes after release
NR by 200
100
30
10
3
1
0
Ungated RL
Critic-gated RL
Threshold RL
21, 16, 15, 20, 19
5, 7, 5, 4, 3
25
1
105, 76, 78, 65, NR
24, 31, 31, 3, 23
22
37


Channel scope matters: show both the gating benefit and the opportunity to change the outcome
Paired-decision effect sizes are conditional diagnostics across sampled states, not causal channel ablations
Polymer: handoff benefit, five paired seeds
SG minus ungated mean reward
Horizon
Weights
Markov
Residual
Combined
Polymer: observed action-choice consequences, seed 7
Mean absolute paired 200-step return difference
standalone
combined
Column: handoff benefit, five paired seeds
Column: observed action-choice consequences, seed 7

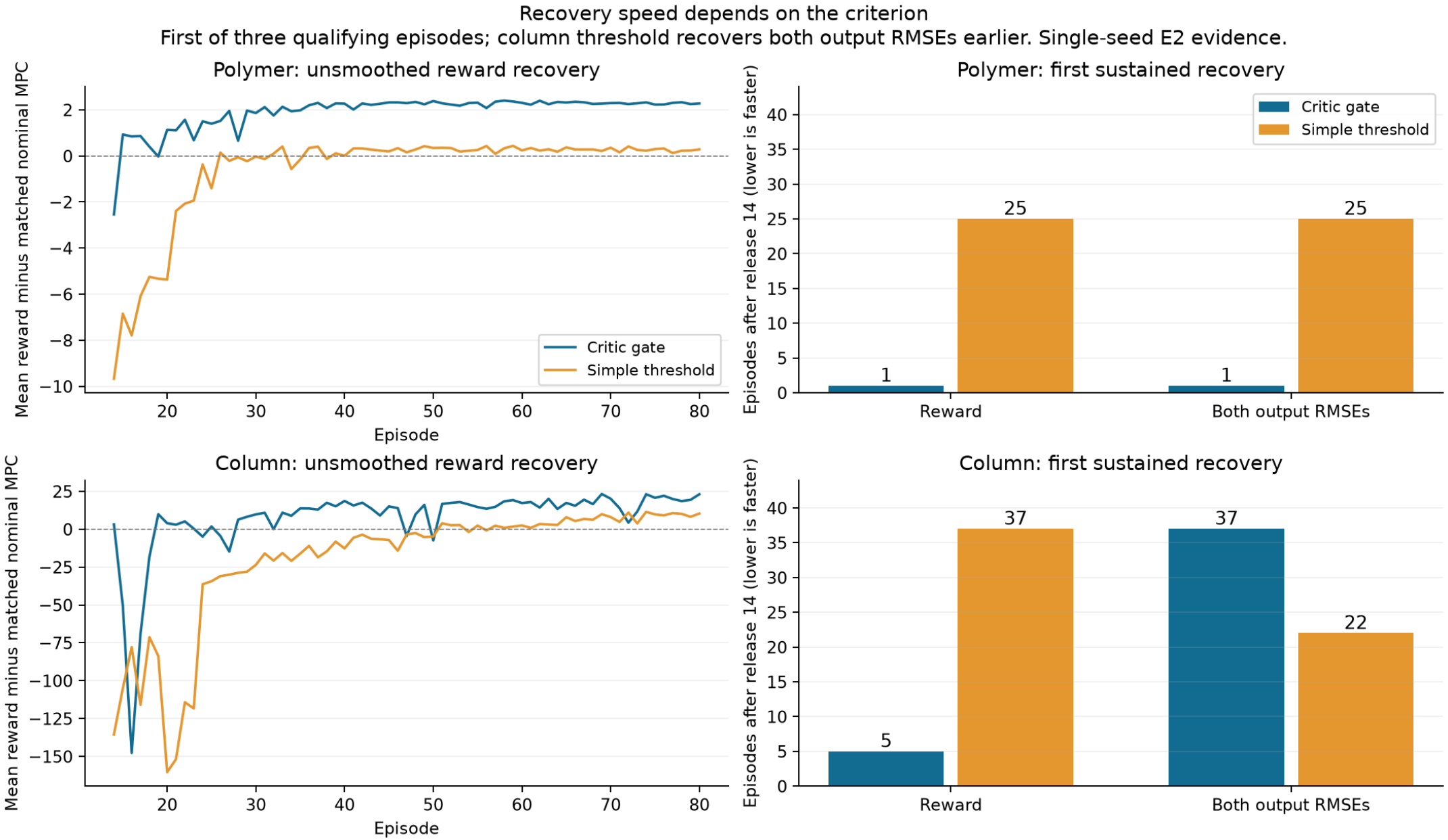

Test of the steady-state explanation: compare only DIFFERENT proposed configurations
PRE-decision history defines steady; first-input tolerance = 0.01% of input identification range
Polymer
Nearly unchanged first MPC input (%)
n=160
n=167
n=192
n=208
Horizon
Weights
Polymer
Median absolute 200-step return difference
n=160
n=167
n=192
n=208
Horizon
Weights
Column
Nearly unchanged first MPC input (%)
n=42
n=23
n=51
n=29
Horizon
Weights
Column
Median absolute 200-step return difference
n=42
n=23
n=51
n=29
Horizon
Weights
Changing / transition
Quasi steady

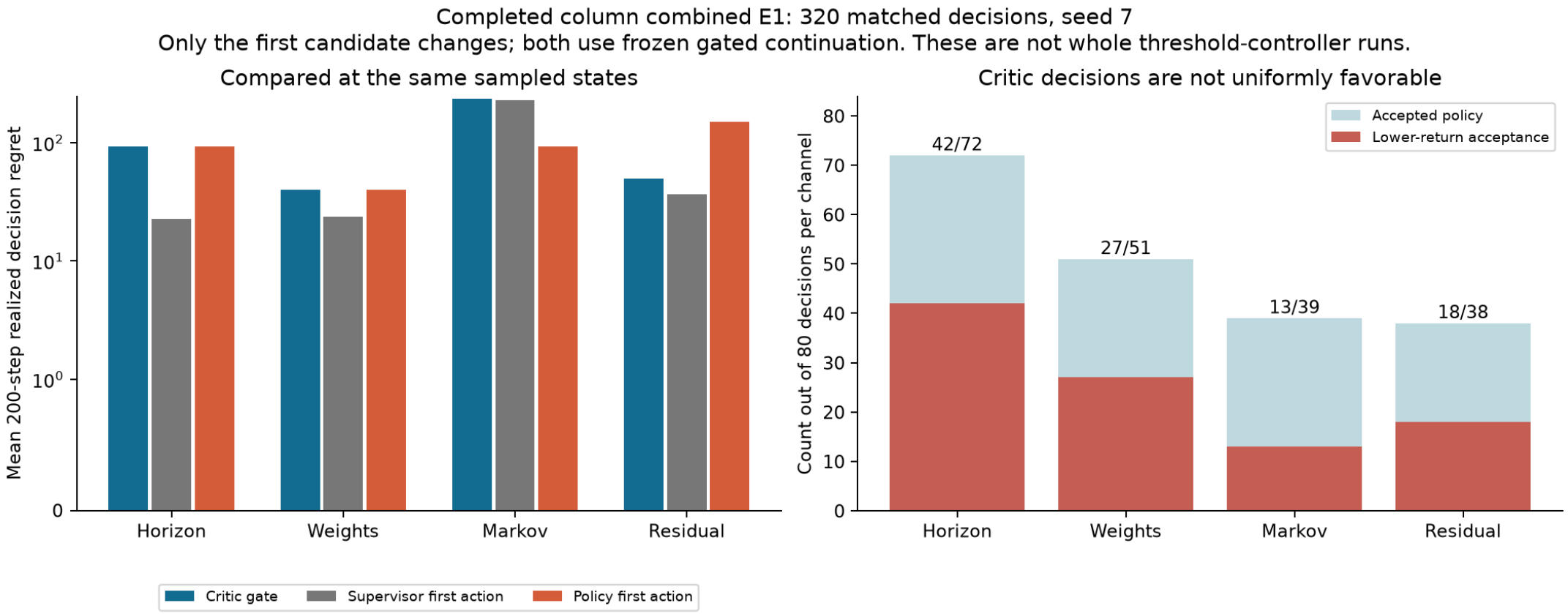

Are the unfavorable horizon/weight decisions simply tiny differences near steady operation?
All lower-return acceptances shown; normalized return differences are sensitivities, not safety probabilities
Polymer: horizon, 116 lower-return acceptances
Polymer: weights, 35 lower-return acceptances
Column: horizon, 32 lower-return acceptances
Column: weights, 24 lower-return acceptances
Changing / transition
Quasi steady
0.5% return sensitivity
Abs return difference / max(1, abs Gsup) (%)
PRE-decision max tracking error / band


Different settings can be reward-equivalent: does the supervisor retain authority?
Immediate reward tie: 0.1% sensitivity; 200-step return tie: 0.5% sensitivity. Same proposals excluded.
Polymer
Different proposals with immediate reward tie (%)
116/160
167/167
150/192
204/208
Horizon
Weights
Polymer
Supervisor selected among 200-step return ties (%)
4/34
27/156
72/109
169/197
Horizon
Weights
Column
Different proposals with immediate reward tie (%)
4/42
9/23
4/51
10/29
Horizon
Weights
Column
Supervisor selected among 200-step return ties (%)
1/13
0/10
7/22
4/22
Horizon
Weights
Changing / transition
Quasi steady

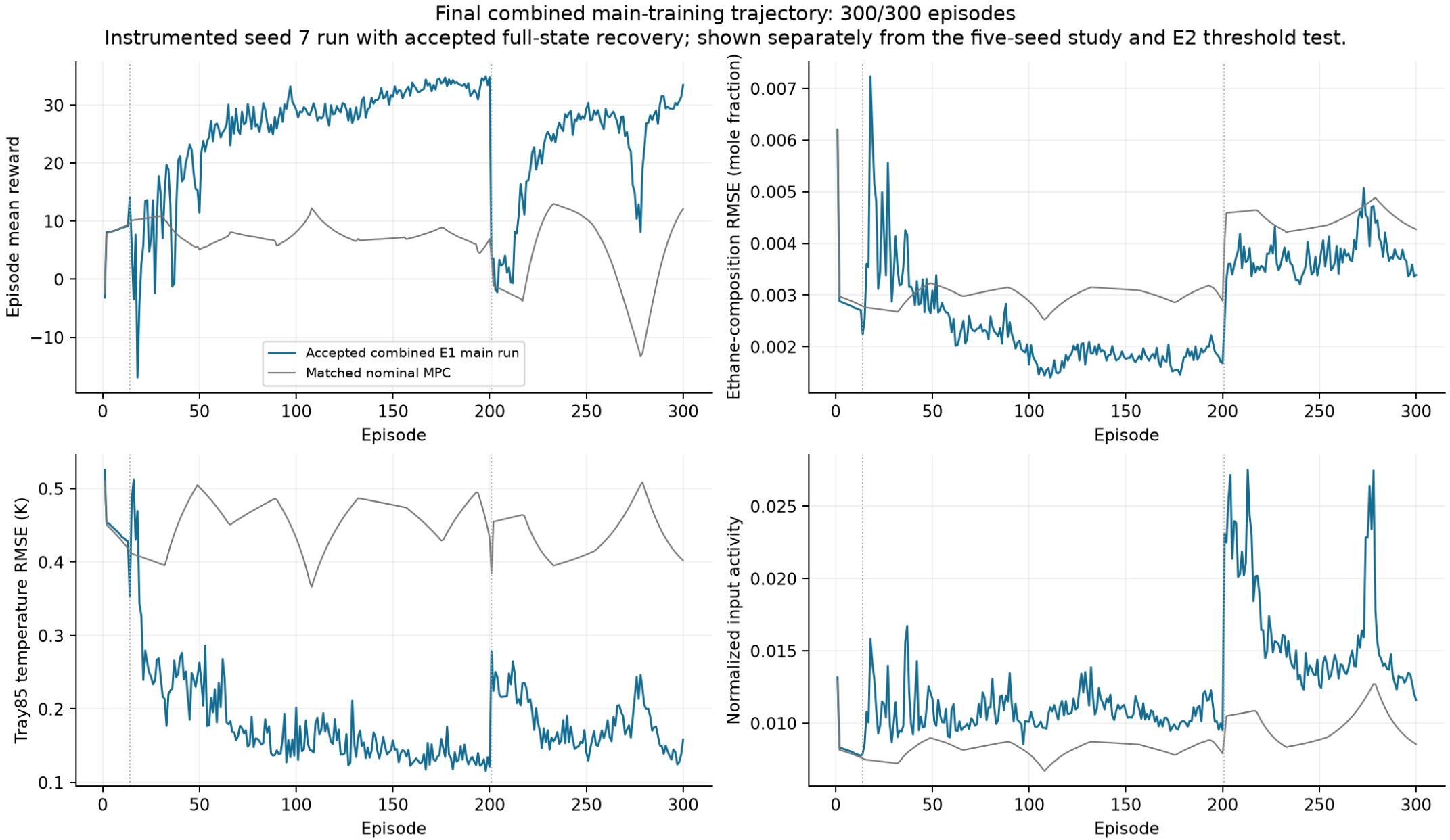

Handoff disruption: critic gate versus simple tracking-error threshold
Episodes 14-20; one E2 realization per method and plant. Tracking bands are not hard safety constraints.
Polymer: tracking error
RMSE / threshold RMSE
Output 1
Output 2
Tracking-band excursions
Steps outside either tracking band (%)
57.1%
99.6%
45.8%
Critic
Threshold
MPC
Input activity (lower is better)
Mean sum abs input change / ID ranges
Column: tracking error
RMSE / threshold RMSE
Output 1
Output 2
Tracking-band excursions
Steps outside either tracking band (%)
51.8%
85.9%
22.7%
Critic
Threshold
MPC
Input activity (lower is better)
Mean sum abs input change / ID ranges
Critic gate
Simple threshold
Nominal MPC

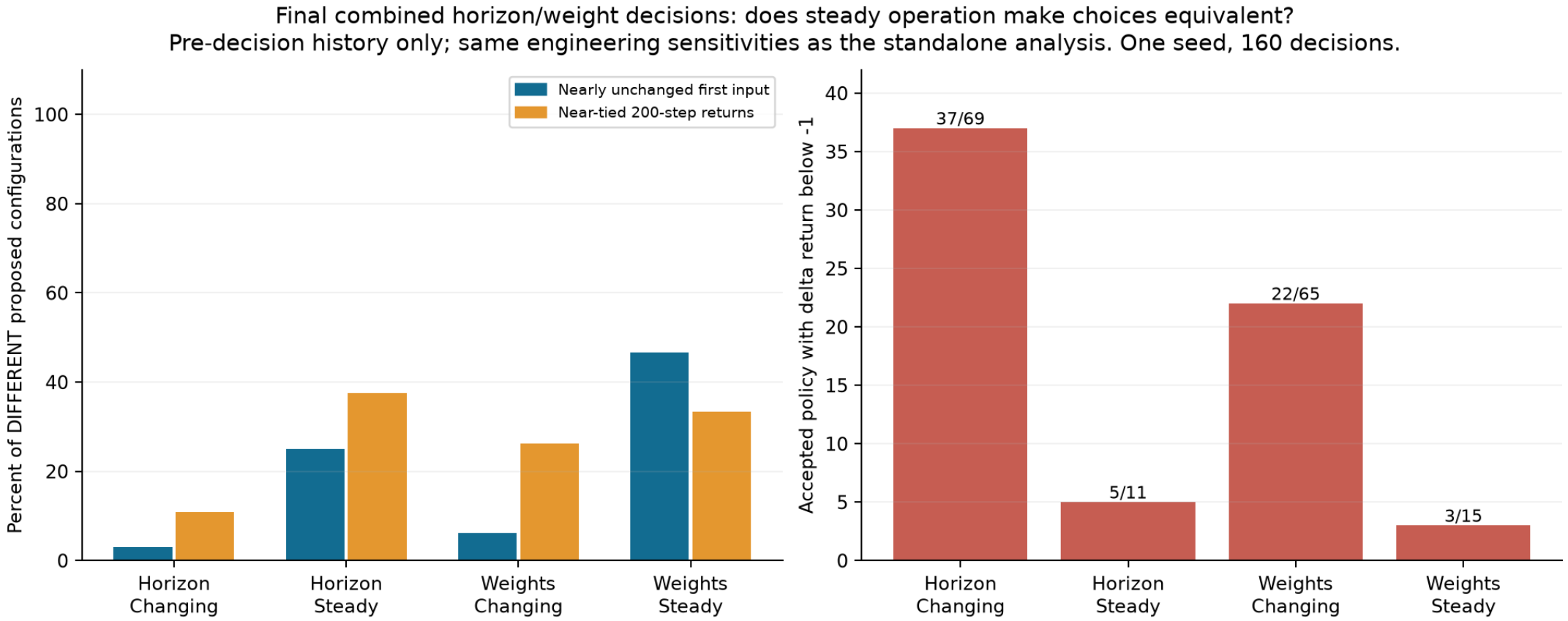
Direct threshold comparison: recovery speed




Completed column combined E1: 320 matched decisions, seed 7
Only the first candidate changes; both use frozen gated continuation. These are not whole threshold-controller runs.
Compared at the same sampled states
Mean 200-step realized decision regret
10^2
10^1
10^0
0
Horizon
Weights
Markov
Residual
Critic gate
Supervisor first action
Policy first action
Critic decisions are not uniformly favorable
Count out of 80 decisions per channel
80
70
60
50
40
30
20
10
0
42/72
27/51
13/39
18/38
Accepted policy
Lower-return acceptance
Horizon
Weights
Markov
Residual

Accepted final combined main-training trajectory




Final combined horizon/weight decisions: does steady operation make choices equivalent?
Pre-decision history only; same engineering sensitivities as the standalone analysis. One seed, 160 decisions.
Nearly unchanged first input
Near-tied 200-step returns
Percent of DIFFERENT proposed configurations
0
20
40
60
80
100
Horizon Changing
Horizon Steady
Weights Changing
Weights Steady
Accepted policy with delta return below -1
0
5
10
15
20
25
30
35
40
37/69
5/11
22/65
3/15
Horizon Changing
Horizon Steady
Weights Changing
Weights Steady